

Ecology Biodiversity and Evolution**Current Final Term Subjective Solved from Internet****1. Vascular and Non Vascular Plants. 10.**

Vascular plants have special tissues called xylem and phloem that form tube- like pathways so that water, minerals, and food can move through plants. Nonvascular plants do not have these special tissues. In nonvascular plants, materials must travel from one cell to the next cell.

Difference Between Vascular and Non-Vascular Plants

Vascular plants or tracheophytes are known for their proper organization of the systems, and bearing flowers, green leaves, stems, roots, woods, and branches, on the contrary, Non-vascular plants or bryophytes do not perfectly fit with these features. Examples of Conifers, Ferns, flowering, and non-flowering plants are examples of vascular plants, while Mosses, Liverworts, and Hornworts are examples of non-vascular plants.

2. Binomial Nomenclature. 5

Binomial nomenclature is the formal naming system for living things that all scientists use. It gives every species a two-part scientific name. For example, a ladybug found in the United States goes by the fancy name of *Harmonia axyridis*. The first part of a scientific name, like *Harmonia*, is called the genus.

3. Ciliophora. 5

The *ciliates* (*phylum Ciliophora*) form a natural group distinguishable from other protozoa by a number of specialized features, including the possession of cilia, which are short hair-like processes, at some stage in their life cycle, the presence of two types of nuclei, and a unique form of sexual reproduction called.

4. Oscillatory and Frequency Selection. 3

The frequency of oscillation is the number of full oscillations in one time unit, say in a second. A pendulum that takes 0.5 seconds to make one full oscillation has a frequency of 1 oscillation per 0.5 second, or 2 oscillations per second

5. Aposematism. 3

Aposematism is the advertising by an animal to potential predators that it is not worth attacking ... by stimulating the evolution of predators to encode the *meaning* of the warning signal, rather than by requiring each new generation to learn the

6. Benefits of Atmosphere.

The Earth's atmosphere protects and sustains the planet's inhabitants by providing warmth and absorbing harmful solar rays. In addition to containing the oxygen and carbon dioxide, which living things need to survive, the atmosphere traps the sun's energy and wards off many of the dangers of space.

7. Trichocyst.

Trichocyst, a structure in the cortex of certain ciliate and flagellate protozoans consisting of a cavity and long, thin threads that can be ejected in response to certain stimuli. *Trichocysts* may be widely distributed over an organism or restricted to certain areas (e.g., tentacles, papillae, around the mouth).

8. Parastism.

In evolutionary ecology, parasitism is a symbiotic relationship between species, where one organism, the parasite, lives on or in another organism, the host, causing it some harm, and is adapted structurally to this way of life.

9. Realized Niche. 2

Realized niche width is a phrase relating to ecology defining the actual space that an organism inhabits and the resources it can access as a result of limiting pressures from other species (e.g. superior competitors).

10. Ozone depleting substances. 2

1. chlorofluorocarbons (CFCs)
2. halon.
3. carbon tetrachloride (CCl₄)
4. methyl chloroform (CH₃CCl₃)
5. hydrobromofluorocarbons (HBFCs)
6. hydrochlorofluorocarbons (HCFCs)
7. methyl bromide (CH₃Br)
8. bromochloromethane (CH₂BrCl)

11. Comparative Anatomy. 2

Comparative anatomy is the study of similarities and differences in the anatomy of different species. It is closely related to evolutionary biology and phylogeny (the evolution of species).

12. Digestive System of Platyhelminthes. 3

Because they don't have a body cavity, flatworms are considered acoelomates. They have simple digestive systems, with mouths to take in food and long digestive tracts to diffuse it around the body. Most flatworms take in food via their mouth, then move it into a digestive gut that attaches to the digestive structures.

13. Characteristics of Mammalians. 5

Mammals have four limbs and produce amniotic eggs. The mammal class is defined by the presence of mammary glands and hair (or fur). Other traits of mammals include sweat glands in their skin, alveoli in their lungs, a four-chambered heart, and a brain covering called the neocortex.

14. Energy flow In Ecosystem. 2

Energy moves life. The cycle of energy is based on the flow of energy through different trophic levels in an ecosystem. Our ecosystem is maintained by the cycling energy and nutrients obtained from different external sources. ... The herbivores at the second trophic level, use the plants as food which gives them energy.

15. Characteristics of chordate. 5

Chordates form a phylum of animals that are defined by having at some stage in their lives all of the following anatomical features:

A notochord, a fairly stiff rod of cartilage that extends along the inside of the body. ...

A dorsal neural tube. ...

Pharyngeal slits. ...

Post-anal tail. ...

An endostyle.

16. Effects Of Acid Rain. 10 Marks.

The ecological effects of acid rain are most clearly seen in aquatic environments, such as streams, lakes, and marshes where it can be harmful to fish and other wildlife. As it flows through the soil, acidic rain water can leach aluminum from soil clay particles and then flow into streams and lakes.

Acid rain has been shown to have adverse impacts on forests, freshwaters, and soils, killing insect and aquatic life-forms, causing paint to peel, corrosion of steel structures such as bridges, and weathering of stone buildings and statues as well as having impacts on human health.

17. Define trophic cascade. 2

Trophic cascades are powerful indirect interactions that can control entire ecosystems. Trophic cascades occur when predators limit the density and/or behavior of their prey and thereby enhance survival of the next lower trophic level.

18. What are herbivores and note on Herbivory. 10 Marks

An herbivore is an animal or insect that only eats vegetation, such as grasses, fruits, leaves, vegetables, roots and bulbs. Herbivores only eat things that need photosynthesis to live. This excludes insects, spiders, fish and other animals.

Herbivory is the consumption of plant material by animals, and herbivores are animals adapted to eat plants. As in predator-prey interactions, this interaction drives adaptations in both the herbivore and the plant species it eats.

19. Discuss in detail evidence of theory of Natural Selection.

In 1859, Charles Darwin set out his theory of evolution by natural selection as an explanation for adaptation and speciation. He defined natural selection as the "principle by which each slight variation [of a trait], if useful, is preserved".

The theory of evolution by natural selection, first formulated in Darwin's book "On the Origin of Species" in 1859, is the process by which organisms change over time as a result of changes in heritable physical or behavioral traits.

20. Define Grazers and Browsers.

This is contrasted with grazing, usually associated with animals feeding on grass or other low vegetation. Alternatively, grazers are animals eating mainly grass, and browsers are animals eating mainly non-grasses, which include both woody and herbaceous dicots

21. Structure of Golgi Apparatus.

The Golgi apparatus, also called Golgi complex or Golgi body, is a membrane-bound organelle found in eukaryotic cells (cells with clearly defined nuclei) that is made up of a series of flattened stacked pouches called cisternae. It is located in the cytoplasm next to the endoplasmic reticulum and near the cell nucleus.

22. Balance Polymorphism with Example. 3

Balanced polymorphism is a situation in which two different versions of a gene are maintained in a population of organisms because individuals carrying both versions are better able to survive than those who have two copies of either version alone. ... Different versions of a gene are called alleles.

23. Reptile Order with example.

Order	Example
Crocodylia: crocodiles, alligators, caimans, gharials	caiman

Sphenodontia: tuataras

tuatara

Squamata: lizards, snakes

lizard

Testudines: turtles, tortoises, terrapins

terrapin

24. Prevention of Animals From Dehydration.

Drink plenty of water, as directed by your doctor. Eat foods with high amounts of water like fruits and vegetables. Avoid or limit drinks with caffeine like coffee, teas and soft drinks. Avoid or limit drinks with alcohol.

25. Aneuploidy and Euploidy.

Changes in chromosome number can occur by the addition of all or part of a chromosome (aneuploidy), the loss of an entire set of chromosomes (monoploidy) or the gain of one or more complete sets of chromosomes (euploidy).

26. Composition of Matrix in Mitochondria.

In the mitochondrion, the matrix is the space within the inner membrane. The mitochondrial matrix contains the mitochondria's DNA, ribosomes, soluble enzymes, small organic molecules, nucleotide cofactors, and inorganic ions.

27. Monocot and Dicot.

Monocots have only one seed leaf inside the seed coat. It is often only a thin leaf, because the endosperm to feed the new plant is not inside the seed leaf. Dicots have two seed leaves inside the seed coat. They are usually rounded and fat, because they contain the endosperm to feed the embryo plant.

28. Adaptedness.

Monocots have only one seed leaf inside the seed coat. It is often only a thin leaf, because the endosperm to feed the new plant is not inside the seed leaf. Dicots have two seed leaves inside the seed coat. They are usually rounded and fat, because they contain the endosperm to feed the embryo plant.

29. Bottleneck Effects.

The bottleneck effect is an extreme example of genetic drift that happens when the size of a population is severely reduced. Events like natural disasters (earthquakes, floods, fires) can decimate a population, killing most individuals and leaving behind a small, random assortment of survivors.

30. Ways of Reproductive Isolation.

These include temporal isolation, ecological isolation, behavioral isolation, and mechanical isolation. Post-zygotic barriers: barriers that come into play after two species have mated. These include genetic incompatibility, zygotic mortality, hybrid inviability, hybrid sterility, and hybrid breakdown.

31. Define Trophic Cascades.

Trophic cascades are powerful indirect interactions that can control entire ecosystems. Trophic cascades occur when predators limit the density and/or behavior of their prey and thereby enhance survival of the next lower trophic level.

32. Primary productivity.

Primary productivity, in ecology, the rate at which energy is converted to organic substances by photosynthetic producers (photoautotrophs), which obtain energy and nutrients by harnessing sunlight, and chemosynthetic producers (chemoautotrophs), which obtain chemical energy through oxidation..

33. Water Cycle.

The cycle of processes by which water circulates between the earth's oceans, atmosphere, and land, involving precipitation as rain and snow, drainage in streams and rivers, and return to the atmosphere by evaporation and transpiration.

Step 1: Evaporation. The **water cycle** starts with evaporation. ...

Step 2: Condensation. ...

Step 3: Sublimation. ...

Step 4: Precipitation. ...

Step 5: Transpiration. ...

Step 6: Runoff. ...

Step 7: Infiltration.

34. Global Warming.

Global warming occurs when carbon dioxide (CO₂) and other air pollutants and greenhouse gases collect in the atmosphere and absorb sunlight and solar radiation that have bounced off the earth's surface.

Global warming, the gradual heating of Earth's surface, oceans and atmosphere, is caused by human activity, primarily the burning of fossil fuels that pump carbon dioxide (CO₂), methane and other greenhouse gases into the atmosphere. ... Already, **global warming** is having a measurable **effect** on the planet.

The **effects of global warming or climate** damage include far-reaching and long-lasting changes to the natural environment, to ecosystems and human societies caused directly or indirectly by human emissions of greenhouse gases. It also includes the economic and social changes which stem from living in a **warmer** world.

35. K selected and R selected.

The two evolutionary "strategies" are termed r-selection, for those species that produce many "cheap" offspring and live in unstable environments and K-selection for those species that produce few "expensive" offspring and live in stable environments.

Typical examples of r-species are mice, rabbits, weeds and bacteria, which have a lot of offspring, but a short life expectancy. Examples of organisms undergoing K-selection are tortoises, elephants, people, and sequoia trees: their offspring are few but long-lived.

36. Homology and Heterology.

Homology is the state of having the same or similar relation, relative position, or structure.

What is heterology: Lack of correspondence between bodily parts, as in structure, arrangement.

37. Thylakoid.

Thylakoids are membrane-bound compartments inside chloroplasts and cyanobacteria. They are the site of the light-dependent reactions of photosynthesis. Thylakoids consist of a thylakoid membrane surrounding a thylakoid lumen

38. Characteristics of Arthropodes.

Characteristics of arthropods include:

A segmented body with a head, a thorax, and abdomen segments.

Appendages on at least one segment. ...

A nervous system.

A hard exoskeleton made of chitin, which gives them physical protection and resistance to drying out.

39. Water Pollutants.

Here are five main types of water pollutants:

MICROBIAL. Pathogens can be bacteria, protozoa or viruses. ...

ORGANIC MATERIAL. Organic materials are things like petroleum, insecticides and herbicides, detergents, disinfecting cleaners, and prescription drugs. ...

INORGANIC MATERIAL. ...
MACROSCOPIC. ...
THERMAL.

40. Molecular Biology Effect Evolution. 10 marks.

Molecular evolution is the process of change in the sequence composition of cellular molecules such as DNA, RNA, and proteins across generations. The field of molecular evolution uses principles of evolutionary biology and population genetics to explain patterns in these changes.

Molecular biology has supported and extended our understanding of evolutionary relationships based on traditional anatomy. Heritability and variation in traits are essential parts of Darwin's theory of evolution by natural selection. ... Molecular biology has clarified the nature of genes and the sources of variation.

Changes that do not affect functionality are called "neutral substitutions" and their accumulation is not affected by natural selection. As a result, neutral substitutions ...

41. Define Gene Drift. 2

Genetic drift describes random fluctuations in the numbers of gene variants in a population. Genetic drift takes place when the occurrence of variant forms of a gene, called alleles, increases and decreases by chance over time. These variations in the presence of alleles are measured as changes in allele frequencies.

42. Characteristics of Mitochondria. 2

They are:

the outer **mitochondrial membrane**,
the intermembrane space (the space between the outer and inner membranes),
the **inner mitochondrial membrane**,
the cristae space (formed by infoldings of the **inner membrane**), and.
the matrix (space within the **inner membrane**).

43. What are the effects of exponential decline of bond energy in a trophic chain.

Exponential decline of chemical-bond energy limits the lengths of trophic chains and the ...
•Bottom-up effects: when effect flows up through a trophic chain

44. Explain the Biogeography.

Biogeography, the study of the geographic distribution of plants, animals, and other forms of life. It is concerned not only with habitation patterns but also with the factors responsible for variations in distribution. Earth's floral regions: Floral kingdoms, subkingdoms, and major regions of the world.

45. Economic Importance of Phylum Myxozoa.

The economically most significant diseases worldwide caused by myxosporeans in cultured fishes are PKD-Proliferative Kidney Disease, caused by a Malacosporea member, *Tetracapsuloides bryosalmonae*, and whirling disease, caused by a Myxosporea member *Myxobolus cerebralis*; both diseases affect salmonids

46. Write Detail note on invasive Species. 5.

An **invasive species** is an organism that causes ecological or economic harm in a new environment where it is not native. ... **Invasive species** are capable of causing extinctions of native plants and animals, reducing biodiversity, competing with native organisms for limited resources, and altering habitats.

47. Note on rate of Evolution. 5.

Rate of evolution is a measurement of the **rate** of genotype change of species and organisms over a period of time.

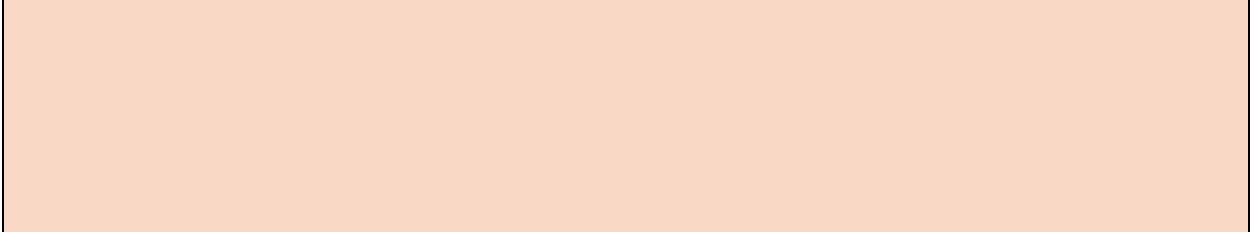
The **rate of evolution** is typically deduced from three factors: mutation **rate**, population size and the fixation probability of new mutations. ... Time to fixation is influenced by population structure, although its effects are complex and at times counterintuitive

48. Write note on phylum Ciliophora. 5.

: a phylum or subphylum of protozoans that possess cilia during some phase of the life cycle and usually have nuclei of two kinds — compare sarcomastigophora.

The ciliates (phylum Ciliophora) form a natural group distinguishable from other protozoa by a number of specialized features, including the possession of cilia, which are short **hair-like** processes, at some stage in their **life cycle**, the presence of two types of nuclei, and a unique form of sexual **reproduction** called

49. Main Affects of Acid rain. 10 Marks.



CONFIDENTIAL